Chapter 1
Introduction

Data Mining: Data mining is the process of extracting useful information from large sets of data. It involves using various techniques from statistics, machine learning, and database systems to identify patterns, relationships, and trends in the data. This information can then be used to make data-driven decisions, solve business problems, and uncover hidden insights.

Data Mining versus Predictive Analytics

Predictive analytics uses historical data and statistical models to forecast future outcomes, while data mining is a foundational process that extracts patterns and insights from large datasets. Their relationship is sequential and symbiotic: data mining finds the patterns, and predictive analytics uses those patterns to build models that make predictions about the future.

Aspect	Data Mining	Predictive Analytics
Primary Purpose	An exploratory, descriptive process focused on uncovering previously unknown patterns, anomalies, and relationships in existing data.	A goal oriented, forecasting process that assesses the probability of future trends and events.
Time Orientation	Past oriented, looking backward at historical data to find hidden insights.	Future oriented, using past data to anticipate what will happen next.
Outcome	Produces insights, summaries, and classifications that describe the data.	Generates predictions, probabilities, and forecasts to guide decision making.
Example Techniques	Techniques include clustering to group similar items, association rule learning to find relationships between variables, and anomaly detection to spot unusual data points.	Method includes regression analysis, machine learning algorithms, time series forecasting.
Role in the Workflow	Often serves as foundational step. The patterns and insights discovered through data mining provides the basis for creating predictive models.	Leverages the insights and patterns that discovered during the data mining phase to build models that project outcomes.

KDD Process in Data Mining: Data mining is also referred as knowledge discovery from data (KDD). Knowledge discovery process is depicted in figure 1.

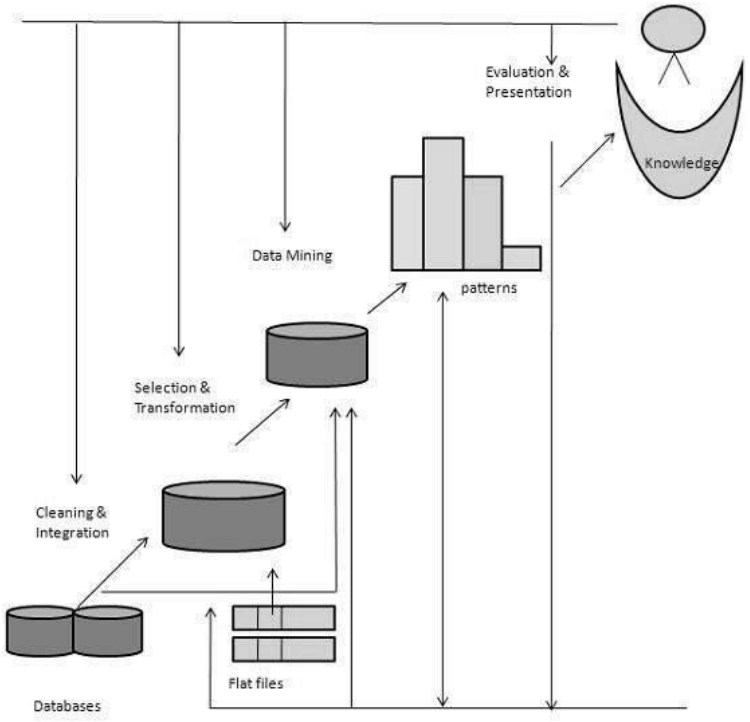


Figure 1: Data mining as a step in the process of knowledge discovery

Knowledge discovery process consists of following steps:

(a) **Data Cleaning** − In this step, the noise and inconsistent data is removed.

(b) **Data Integration** − In this step, multiple data sources are combined.

(c) **Data Selection** − In this step, data relevant to the analysis task are retrieved from the database.

(d) **Data Transformation** − In this step, data is transformed or consolidated into forms appropriate for mining by performing summary or aggregation operations.

(e) **Data Mining** − In this step, intelligent methods are applied in order to extract data patterns.

(f) **Pattern Evaluation** − In this step, data patterns are evaluated.

(g) **Knowledge Presentation** − In this step, knowledge is represented.

Architecture of Data Mining System: Architecture of a typical data mining system is provided in figure 2.

A detailed description of parts of data mining architecture is given below:

(a) **Data Sources:** Database, World Wide Web (WWW) and data warehouse are parts of data sources. The data in these sources may be in the form of plain text, spreadsheets, or other forms of media like photos or videos. WWW is one of the biggest sources of data.

(b) **Database Server:** The database server contains the actual data ready to be processed. It performs the task of handling data retrieval as per the request of the user.

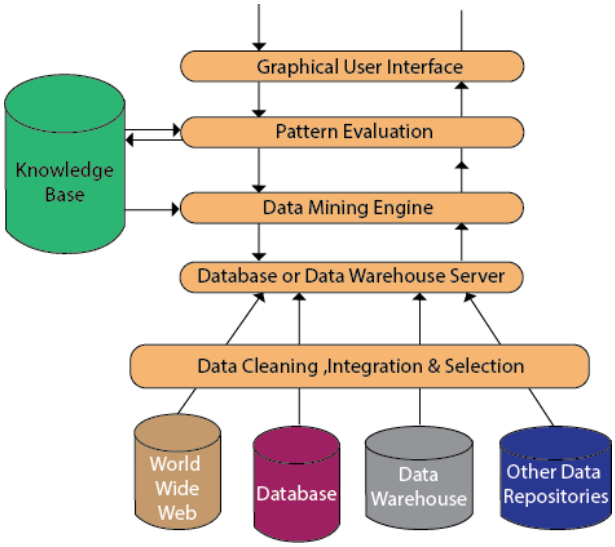


Figure 2: Architecture of a typical Data Mining

(c) **Data Mining Engine:** It is one of the core components of the data mining architecture that performs all kinds of data mining techniques like association, classification, characterization, clustering, prediction, etc.

(d) **Pattern Evaluation Modules:** They are responsible for finding interesting patterns in the data and sometimes they also interact with the database servers for producing the result of the user requests.

(e) **Graphic User Interface:** Since the user cannot fully understand the complexity of the data mining process so graphical user interface helps the user to communicate effectively with the data mining system.

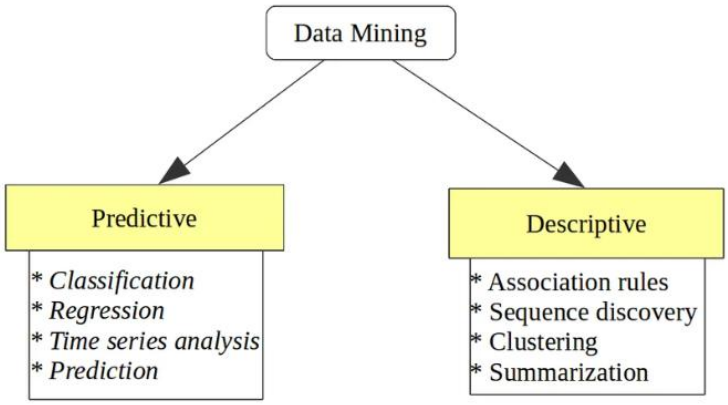
(f) **Knowledge Base:** Knowledge Base is an important part of the data mining engine that is quite beneficial in guiding the search for the result patterns. Data mining engines may also sometimes get inputs from the knowledge base. This knowledge base may contain data from user experiences. The objective of the knowledge base is to make the result more accurate and reliable.

What Kinds of data can be mined?

- **Relational Databases** − A database system is also called a database management system. It includes a set of interrelated data, called a database, and a set of software programs to handle and access the data.
- **Data warehouse** – A data warehouse is a repository of information collected from multiple sources, stored under a unified schema, and usually residing at a single site. Data warehouses are constructed via a process of data cleaning, data integration, data transformation, data loading, and periodic data refreshing.
- **Transactional Databases** − A transactional database includes a file where each record defines a transaction. A transaction generally contains a unique transaction identity number (trans ID) and a list of the items creating up the transaction (such as items purchased in a store).

- **Object-Relational Databases** − Object-relational databases are assembled based on an object-relational data model. This model continues the relational model by supporting a rich data type for managing complex objects and object orientation.
- **Temporal Database** − A temporal database generally stores relational data that contains time-related attributes. These attributes can include multiple timestamps, each having several semantics.
- **Sequence Database** − A sequence database stores sequences of ordered events, with or without a factual idea of time. For example, customer shopping sequences, Web click streams, and biological sequences.
- **Time-Series Database** − A time-series database stores sequences of values or events accessed over repeated measurements of time (e.g., hourly, daily, weekly). An example includes data collected from the stock exchange, stock control, and the measurement of natural phenomena (like temperature and wind).

Tasks accomplished by Data Mining: Data Mining tasks involve finding patterns and useful information in large datasets.



Predictive Data Mining: It is a data mining technique that uses historical data and machine learning or statistical algorithm to forecast future events, trends, and outcomes.

Descriptive Data Mining: It focuses on uncovering patterns, trends, and relationships within existing data. This category of data mining doesn't aim to make predictions but rather seeks to provide valuable insights into historical or current data.

Classification: Classification is a technique in data mining that involves categorizing or classifying data objects into predefined classes, categories, or groups based on their features or attributes. Classification is a supervised learning technique that used labelled data to build a model that can predict the class of new, unseen data.

Regression: Regression is a supervised learning technique used to predict a continuous numerical value by analysing the relationship between a dependent (target) variable and one or more independent (predictor) variables using past data.

Prediction: Prediction utilizes historical data to forecast unknown or future values. Prediction aims to estimate the value of a target variable, which can be either continuous (regression) or categorical (classification).

Time Series Analysis: Time series analysis is a way to analyse a sequence of data points collected over an interval of time. It focuses on revealing underlying patterns such as trends, seasonality, and cycles, that are not apparent in static dataset

Clustering: Clustering is a technique used to group similar data points together based on their features and characteristics. It is unsupervised learning technique.

Summarization: Summarization is the process of condensing long texts, documents, or data into a concise version while retaining core messages and key points. It involves identifying crucial information to create a shorter, actionable summary.

Association: Association rule mining finds interesting associations and relationships among large sets of data items. This rule shows how frequently an itemset occurs in a transaction.

Sequence Discovery: It is a data mining technique used to identify frequently occurring ordered patterns in sequential data, such as a customer's purchase history or web user activity.

Data Mining Applications: Large number of applications are using data mining concept. Some of them are depicted in Figure 3.

(a) **Scientific Analysis:** Scientific simulations are generating bulks of data every day. This includes data collected from nuclear laboratories, data about human psychology, etc. Data mining techniques are capable of the analysis of these data. Now we can capture and store more new data faster than we can analyse the old data already accumulated. Example of scientific analysis:

- Sequence analysis in bioinformatics
- Classification of astronomical objects
- Medical decision support

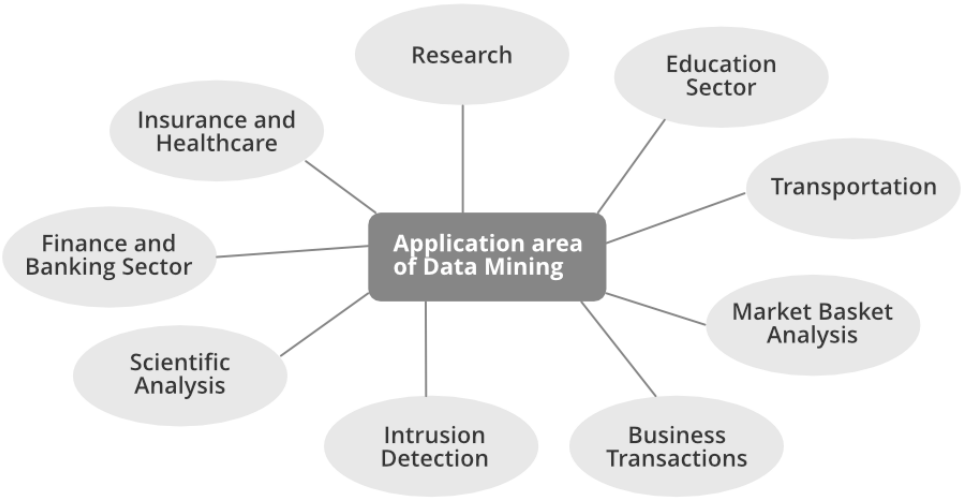


Figure 3: Data Mining Applications

(b) **Intrusion Detection:** A network intrusion refers to any unauthorized activity on a digital network. Network intrusions often involve stealing valuable network resources. Data mining technique plays a vital role in searching intrusion detection, network attacks, and

anomalies. These techniques help in selecting and refining useful and relevant information from large data sets. Data mining technique helps in classify relevant data for Intrusion Detection System. Intrusion Detection system generates alarms for the network traffic about the foreign invasions in the system. Example are:

- Detect security violations
- Misuse Detection
- Anomaly Detection

(c) **Business Transactions:** Every business industry is memorized for perpetuity. Such transactions are usually time-related and can be inter-business deals or intra-business operations. The effective and in-time use of the data in a reasonable time frame for competitive decision-making is definitely the most important problem to solve for businesses that struggle to survive in a highly competitive world. Data mining helps to analyse these business transactions and identify marketing approaches and decision-making. Examples are:

- Direct mail targeting
- Stock trading
- Customer segmentation
- Churn prediction (Churn prediction is one of the most popular Big Data use cases in business)

(d) **Market Basket Analysis:** Market Basket Analysis is a technique that gives the careful study of purchases done by a customer in a supermarket. This concept identifies the pattern of frequent purchase items by customers. This analysis can help to promote deals, offers, sale by the companies and data mining techniques helps to achieve this analysis task. Examples are:

- Data mining concepts are in use for Sales and marketing to provide better customer service, to improve cross-selling opportunities, to increase direct mail response rates.
- Customer Retention in the form of pattern identification and prediction of likely defections is possible by Data mining.
- Risk Assessment and Fraud area also use the data-mining concept for identifying inappropriate or unusual behaviour etc.

(e) **Education:** For analysing the education sector, data mining uses Educational Data Mining (EDM) method. This method generates patterns that can be used both by learners and educators. By using data mining EDM we can perform some educational tasks:

- Predicting students' admission in higher education
- Predicting students profiling
- Predicting student performance
- Teachers teaching performance
- Curriculum development
- Predicting student placement opportunities

(f) **Research:** A data mining technique can perform predictions, classification, clustering, associations, and grouping of data with perfection in the research area. Rules generated by data mining are unique to find results. In most of the technical research in data mining, we create a training model and testing model. The training/testing model is a strategy to measure the precision of the proposed model. It is called Train/Test because we split the

data set into two sets: a training data set and a testing data set. A training data set used to design the training model whereas testing data set is used in the testing model. Examples are:

- Classification of uncertain data.
- Information-based clustering.
- Decision support system
- Web Mining
- Domain-driven data mining
- IoT (Internet of Things) and Cybersecurity
- Smart farming IoT (Internet of Things)

(g) **Healthcare and Insurance:** A Pharmaceutical sector can examine its new deals force activity and their outcomes to improve the focusing of high-value physicians and figure out which promoting activities will have the best effect in the following upcoming months, Whereas the Insurance sector, data mining can help to predict which customers will buy new policies, identify behaviour patterns of risky customers and identify fraudulent behaviour of customers. Examples are:

- Claims analysis i.e. which medical procedures are claimed together
- Identify successful medical therapies for different illnesses
- Characterizes patient behaviour to predict office visits

(h) **Transportation:** A diversified transportation company with a large direct sales force can apply data mining to identify the best prospects for its services. A large consumer merchandise organization can apply information mining to improve its business cycle to retailers. Examples are:

- Determine the distribution schedules among outlets
- Analyse loading patterns

(i) **Financial/Banking Sector:** A credit card company can leverage its vast warehouse of customer transaction data to identify customers most likely to be interested in a new credit product.

- Credit card fraud detection
- Identify 'Loyal' customers
- Extraction of information related to customers
- Determine credit card spending by customer groups

Cross Industry Standard Process for Data Mining (CRISP-DM)

The Cross-industry standard process for data mining, known as CRISP-DM, is an open standard process model that describes common approaches used by data mining experts. It is the most widely-used analytics model. It consists of six phases: Business Understanding, Data Understanding, Data Preparation, Modelling, Evaluation, and Deployment. The process is iterative, allowing for loops and backtracking to previous steps as needed, and is a flexible, practical methodology for solving business problems with data.

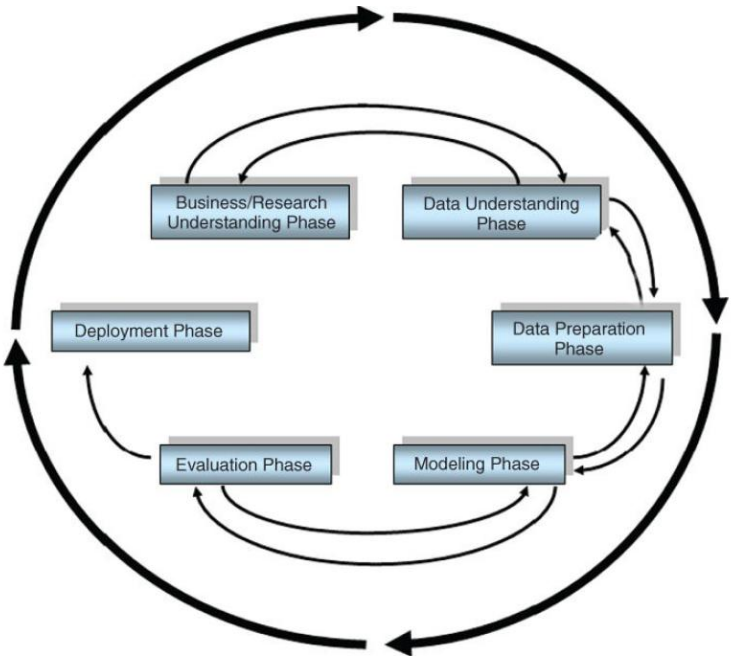


Figure 4: CRISP-DM: Interactive and Adaptive Process

Business/Research Understanding Phase:

a. First, clearly enunciate the project objectives and requirements in terms of the business or research unit as a whole.

b. Then, translate these goals and restrictions into the formulation of a data mining problem definition.

c. Finally, prepare a preliminary strategy for achieving these objectives.

2. **Data Understanding Phase**

a. First, collect the data.

b. Then, use exploratory data analysis to familiarize yourself with the data, and discover initial insights.

c. Evaluate the quality of the data.

d. Finally, if desired, select interesting subsets that may contain actionable patterns.

3. **Data Preparation Phase**

a. This labour-intensive phase covers all aspects of preparing the final data set, which shall be used for subsequent phases, from the initial, raw, dirty data.

b. Select the cases and variables you want to analyse, and that are appropriate for your analysis.

c. Perform transformations on certain variables, if needed.

d. Clean the raw data so that it is ready for the modelling tools.

4. **Modelling Phase**

a. Select and apply appropriate modelling techniques.

b. Calibrate model settings to optimize results.

c. Often, several different techniques may be applied for the same data mining problem.

d. May require looping back to data preparation phase, in order to bring the form of the data into line with the specific requirements of a particular data mining technique.

5. **Evaluation Phase**

a. The modelling phase has delivered one or more models. These models must be evaluated for quality and effectiveness, before we deploy them for use in the field.

b. Also, determine whether the model in fact achieves the objectives set for it in phase 1.

c. Establish whether some important facet of the business or research problem has not been sufficiently accounted for.

d. Finally, come to a decision regarding the use of the data mining results.

6. **Deployment Phase**

a. Model creation does not signify the completion of the project. Need to make use of created models.

b. Example of a simple deployment: Generate a report.

c. Example of a more complex deployment: Implement a parallel data mining process in another department.

d. For businesses, the customer often carries out the deployment based on your model.

Fallacies of Data Mining: It describes some of the limitations of data mining and serves, by implication, as warning that one must approach these techniques with a knowledge of mathematics and statistics.

Fallacy 1. There are data mining tools that we can turn loose on our data repositories, and find answers to our problems.

Reality. No, data mining is not automatic, rather it is a process. Data should be pre-processed, cleansed, properly classified, and missing data must be handled.

Fallacy 2. The data mining process is autonomous, requiring little or no human oversight.

Reality. Data mining is not magic. Without skilled human supervision, blind use of data mining software will only provide you with the wrong answer to the wrong question applied to the wrong type of data. Further, the wrong analysis is worse than no analysis, because it leads to policy recommendations that will probably turn out to be expensive failures. Even after the model is deployed, the introduction of new data often requires an updating of the model. Continuous quality monitoring and other evaluative measures must be assessed, by human analysts.

Fallacy 3. Data mining pays for itself quite quickly.

Reality. It may take time to realize the benefits of data mining. Implementing a response can be expensive itself, in addition to the expense of the collection and storage of data, and the expense of the data mining (purchasing software and hiring of data analysts).

Fallacy 4. Data mining software packages are intuitive and easy to use.

Reality. No, one must understand the statistical and mathematical assumptions of the model which the software uses. Data analysts must combine subject matter knowledge with an analytical mind, and a familiarity with the overall business or research model.

Fallacy 5. Data mining will identify the causes of our business or research problems.

Reality. The knowledge discovery process will help you to uncover patterns of behaviour. Again, it is up to the humans to identify the causes.

Fallacy 6. Data mining will automatically clean up our messy database.

Reality. No, data usually needs to be pre-processed before software is applied to it. This is particularly relevant when considering old data which may require updates which take into consideration changing trends, preferences, and technology.

Fallacy 7. Data mining always provides positive results.

Reality. This is not guaranteed. If used properly, data mining can potentially provide actionable results.
